

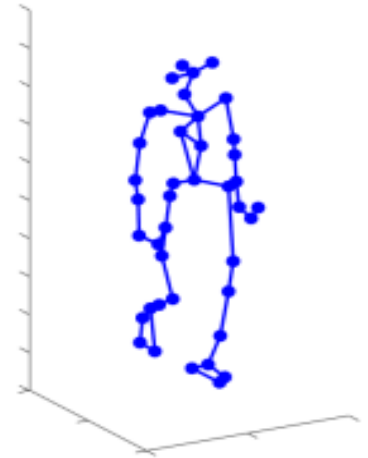
Human Motion Analysis Based on Temporal Variations of Shape Subspaces

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研究背景

■ Motion Analysis Using Motion Capture Data

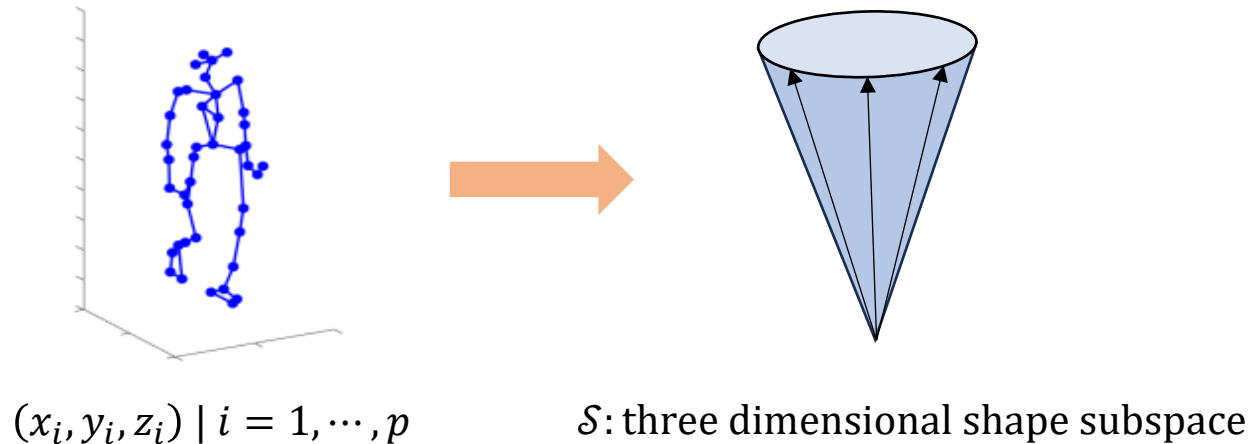
- Motion analysis using motion capture data — a set of key points obtained from sensors attached to the body
- independent of body shape and background information
- applicable to healthcare, sports, and activity recognition



研究背景

■ Shape Subspace

- A three-dimensional subspace of a p -dimensional vector space representing the shapes of a set of p landmarks
- Invariant under linear affine transformations
- Viewpoint-independent



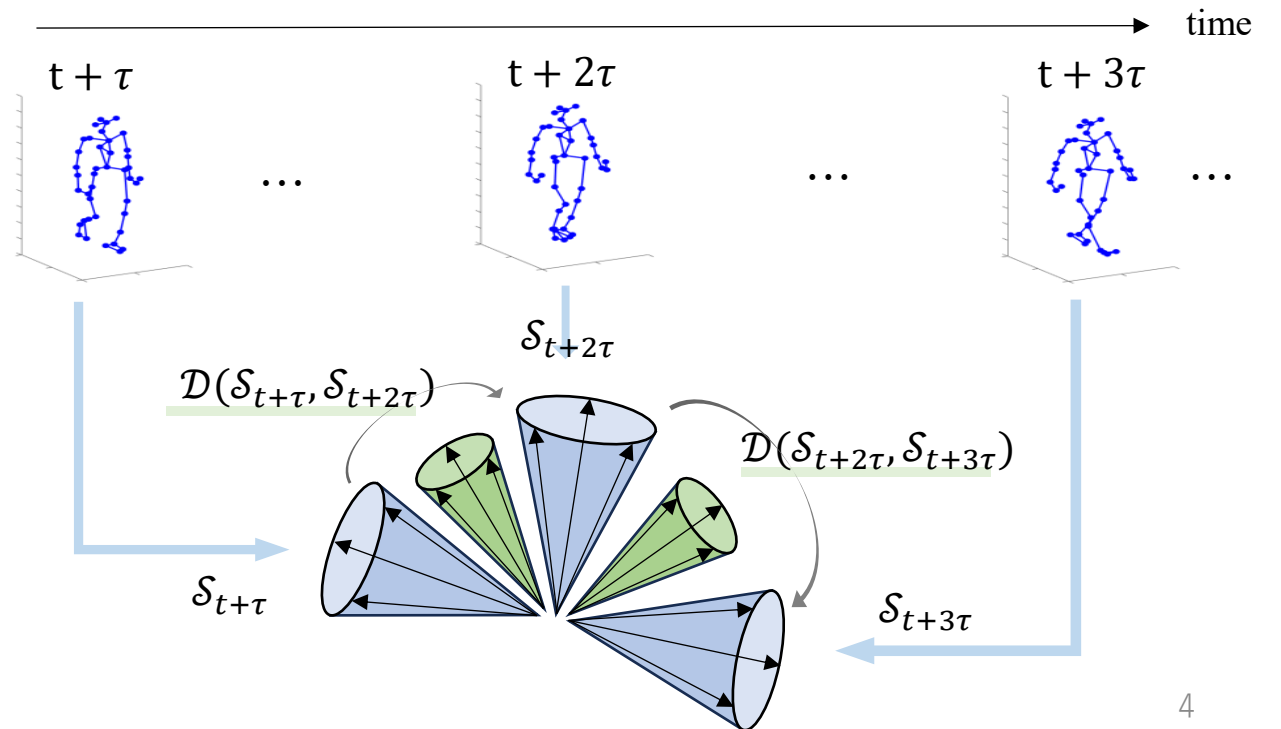
研究背景

■ Motion Analysis Using Shape Subspaces

- Human motion cannot be captured by linear transformations; the shape subspace changes according to the motion
- Each frame's set of key points is represented as a single shape subspace, and the differences between shape subspaces are computed
- Geometrically computing motion variations

$\mathcal{S}_t$: フレーム t における3次元形状部分空間

$\mathcal{D}(\mathcal{S}_{t+\tau}, \mathcal{S}_{t+2\tau})$: $\mathcal{S}_{t+\tau}$ と $\mathcal{S}_{t+2\tau}$ 間の3次元差分部分空間



研究背景

■ Objective

- Motion analysis that geometrically captures essential shape changes, in contrast to directly computing coordinate differences
- In particular, evaluating the periodicity and symmetry of gait is effective for diagnosing gait abnormalities
- Primarily, motion analysis focusing on the periodicity and symmetry of gait is conducted to contribute to the medical field

手法・関連研究

■ Shape Subspace $\mathcal{S}$

$$1. \mathbf{X} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_p)^T = \begin{pmatrix} x_1 & y_1 & z_1 \\ \vdots & \vdots & \vdots \\ x_p & y_p & z_p \end{pmatrix}$$

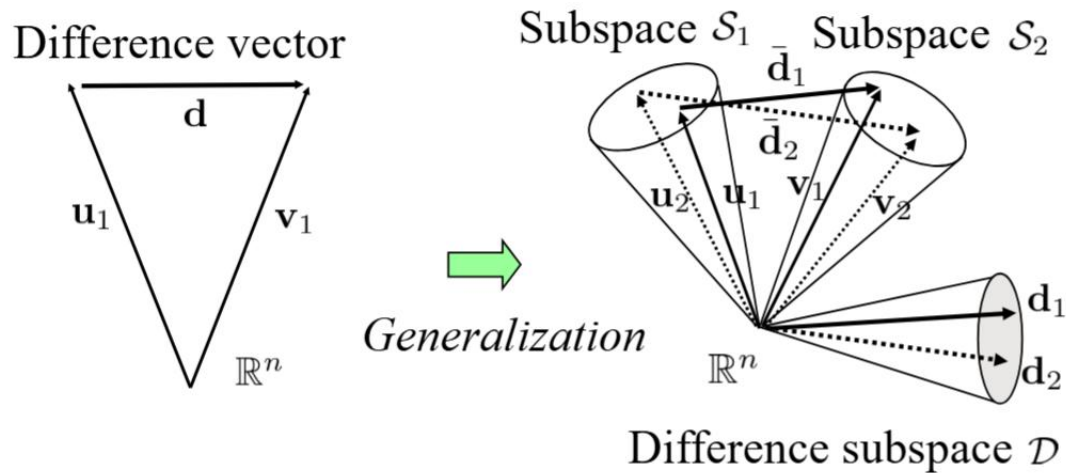
$$2. \mathbf{X}_c = \mathbf{X} - \frac{1}{p} \sum_{i=1}^p \mathbf{r}_i$$

$$3. \mathbf{X}_c = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

The shape subspace $\mathcal{S}$ is defined as the space spanned by the first three column vectors of $\mathbf{U}$

手法・関連研究

■ difference subspace $\mathcal{D}(\mathcal{S}_1, \mathcal{S}_2)$



- Represents the difference between subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$
- The subspace remaining after removing the common subspace from the sum of $\mathcal{S}_1$ and $\mathcal{S}_2$.
- Magnitude of the difference subspace

$$Mag(\mathcal{D}(\mathcal{S}_1, \mathcal{S}_2)) = \sum_{i=1}^d \bar{d}_i^2$$

実験

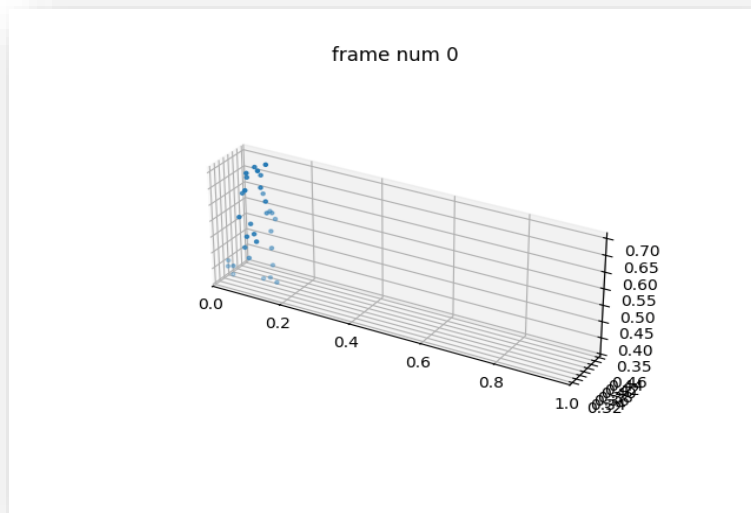
- Targeting the **gait motions** of healthy individuals and patients (stroke patients), we evaluate **symmetry** and **periodicity**

実験

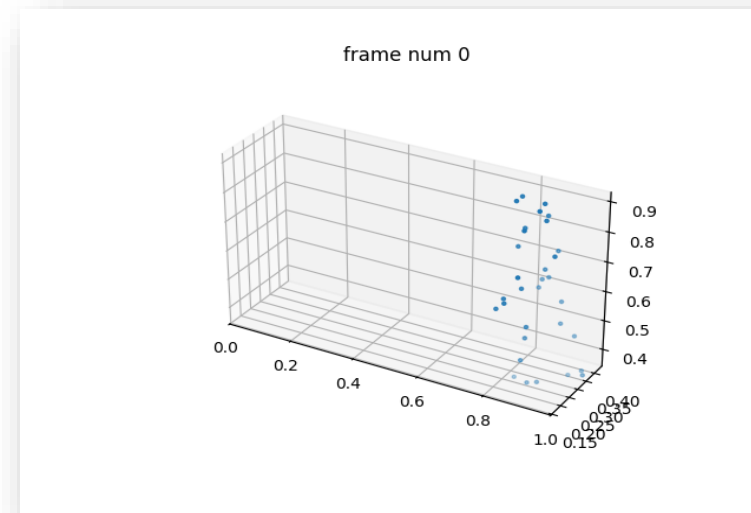
■ dataset

Full-body gait dataset of 138 healthy adults and 50 stroke survivors

Gait motion capture data from 138 healthy adults (aged 21–86) and 50 stroke survivors were used (multiple gait data per individual). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692332/>



healthy

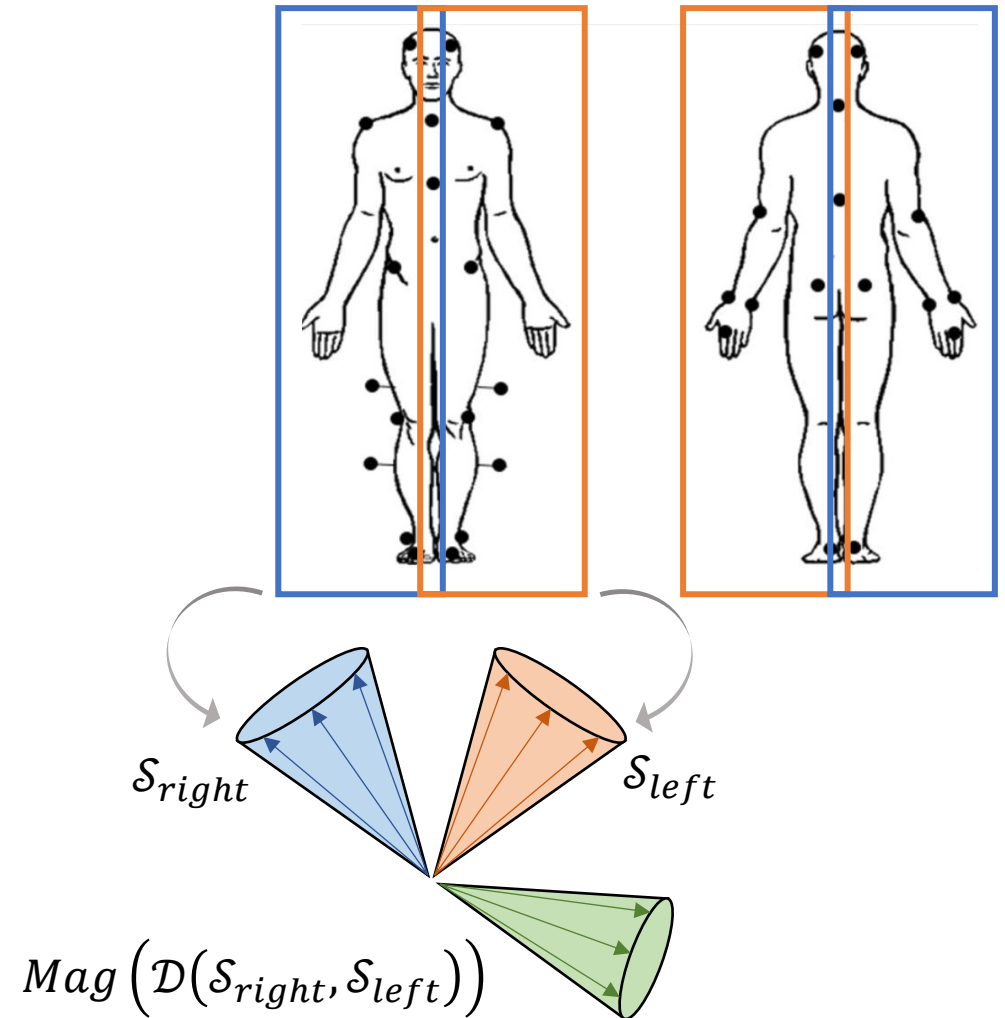


stroke

実験

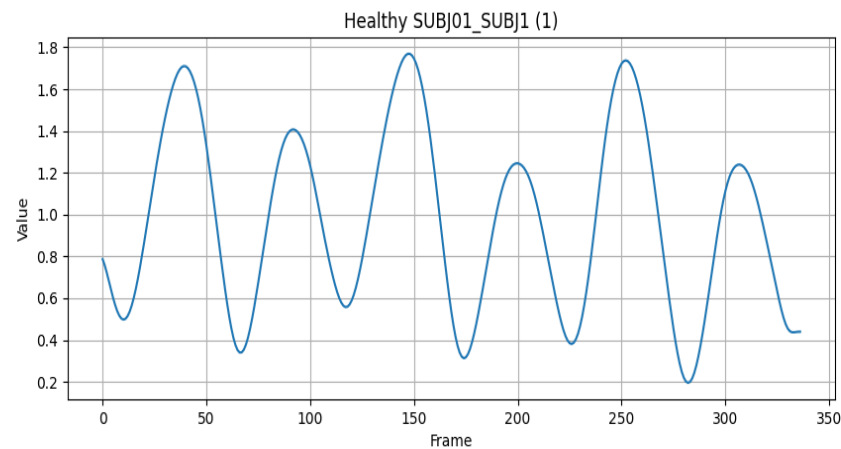
■ Evaluation of Symmetry

1. Key points are divided into right and left sides (central points such as the back and chest are included in both)
2. Shape subspaces are created for the right and left sides
3. For each frame, the magnitude of the difference subspace between shape subspaces is computed

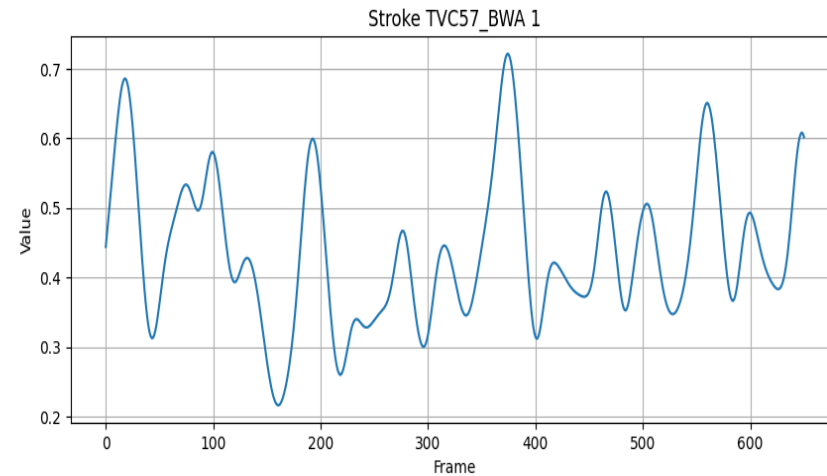


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■ Evaluation of Symmetry (Qualitative Assessment)



healthy



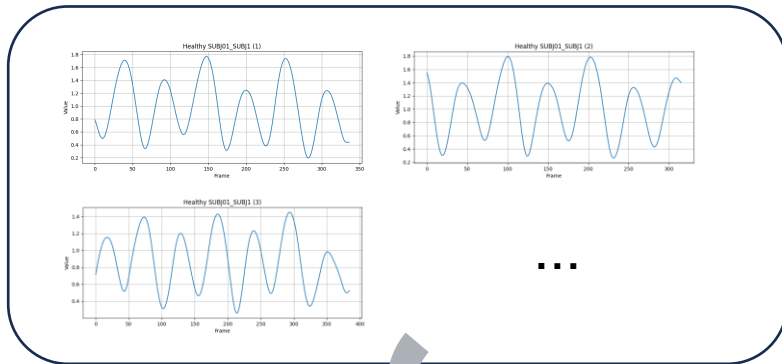
stroke

- It was confirmed that healthy individuals exhibit waveforms with nearly perfect periodicity
- For patients, those with awkward gait tend to show irregularities in the waveforms, whereas some who can walk relatively normally exhibit clean periodicity similar to healthy individuals

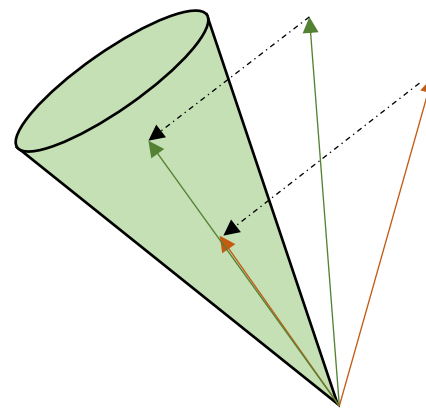
実験

■ Evaluation of Symmetry (Qualitative Assessment)

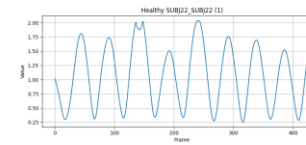
Healthy (training)



Apply PCA to FFT features to construct the normal subspace

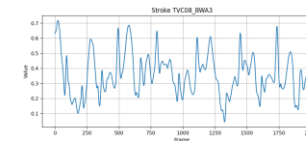


Healthy (testing)



FFT特徴量

Patients



FFT特徴量

The degree of normality is calculated based on the length of FFT feature vectors projected onto the normal subspace. A longer projected vector indicates a higher level of normality

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■ Evaluation of Symmetry

To compare the method and accuracy of shape subspace representation, other methods are also tested

- Shape Subspace

$$mag = Mag(\mathcal{D}(\mathcal{S}_{right}, \mathcal{S}_{left}))$$

- Difference of Mean Vectors

$$mag = \|mean\{(x, y, z)_{right}\} - mean\{(x, y, z)_{left}\}\|$$

- After applying ICP

$$mag = \text{Point Matching Rate}/rmse$$

- Euclidean distance (requires the 2 points clouds to have the same number of points)

$$mag = mean\|\{(x, y, z)_{right} - (x, y, z)_{left}\}\|$$

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■ Evaluation of Symmetry

- Shape Subspace

$$mag = Mag(\mathcal{D}(\mathcal{S}_{right}, \mathcal{S}_{left}))$$

- Difference of Mean Vectors

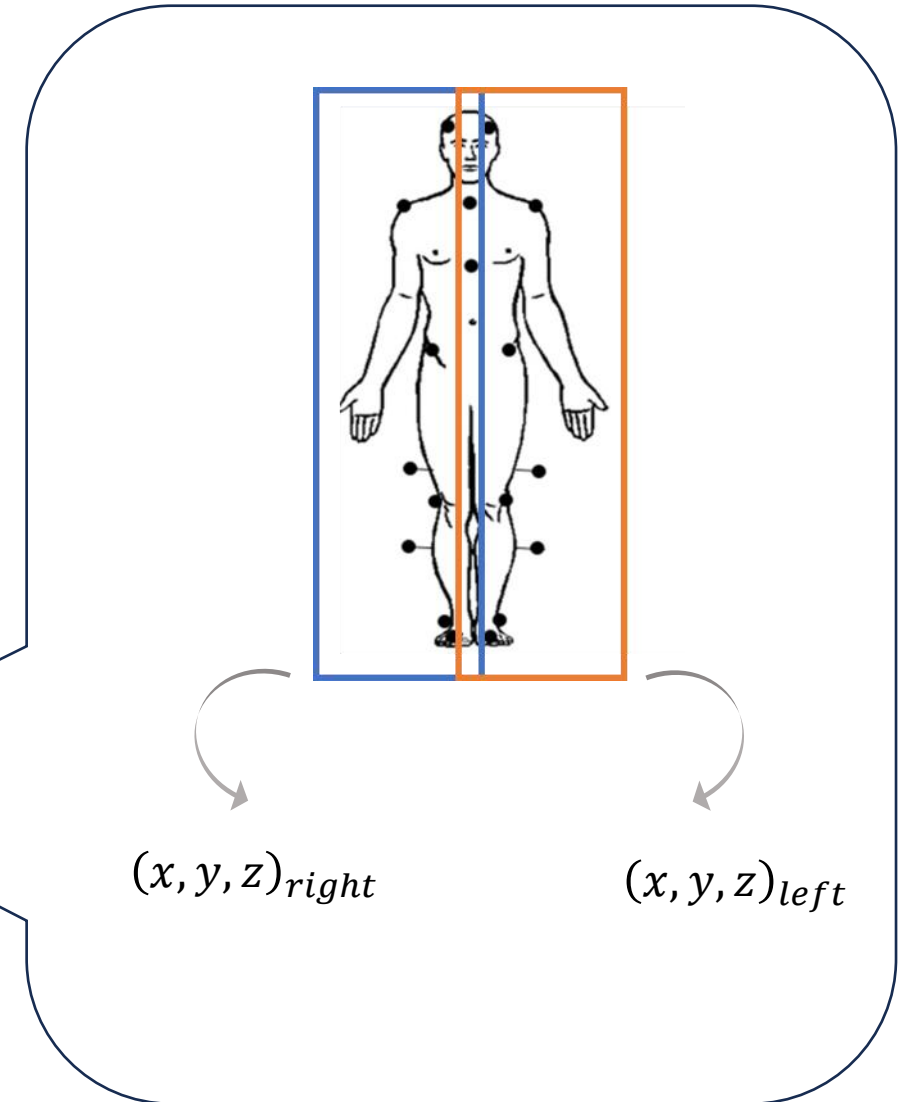
$$mag = \|\text{mean}\{(x, y, z)_{right}\} - \text{mean}\{(x, y, z)_{left}\}\|$$

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- Euclidean distance (requires the 2 points clouds to have the same number of points)

$$mag = \text{mean}\|\{(x, y, z)_{right} - (x, y, z)_{left}\}\|$$

Iterative Closest Point (ICP)

Method for registering 2 points clouds to achieve optimal alignment

1. For each point in the moving point cloud, find the nearest point in the reference point cloud
2. Compute the rotation and translation that minimize the distances between corresponding points
3. Update the alignment and repeat until convergence

After convergence, compute using point correspondence ratio and RMSE

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■ Evaluation of Symmetry

- Shape Subspace

$$mag = Mag(\mathcal{D}(\mathcal{S}_{right}, \mathcal{S}_{left}))$$

- Difference of Mean Vectors

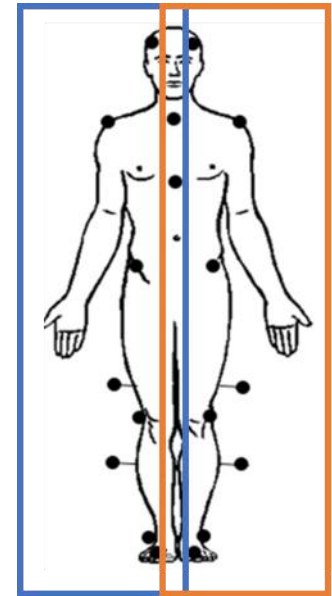
$$mag = \|\text{mean}\{(x, y, z)_{right}\} - \text{mean}\{(x, y, z)_{left}\}\|$$

- After applying ICP

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- Euclidean distance (requires the 2 points clouds to have the same number of points)

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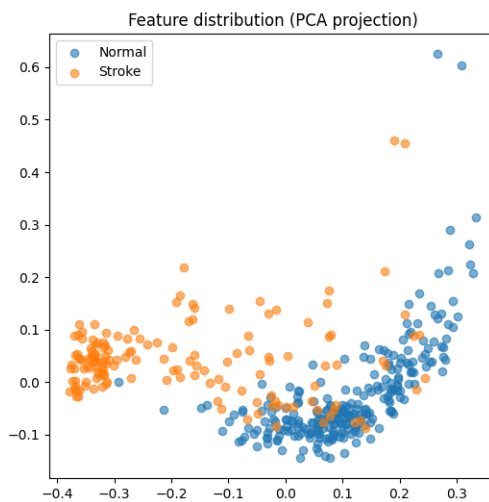
Euclidean distance

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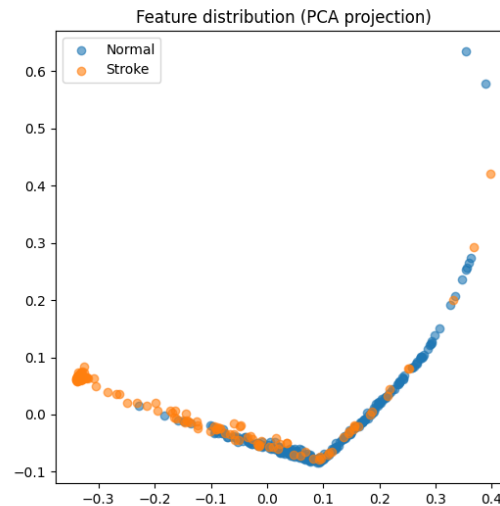
■ Evaluation of Symmetry

- Distribution and separability (Fisher's discriminant ratio) of FFT features

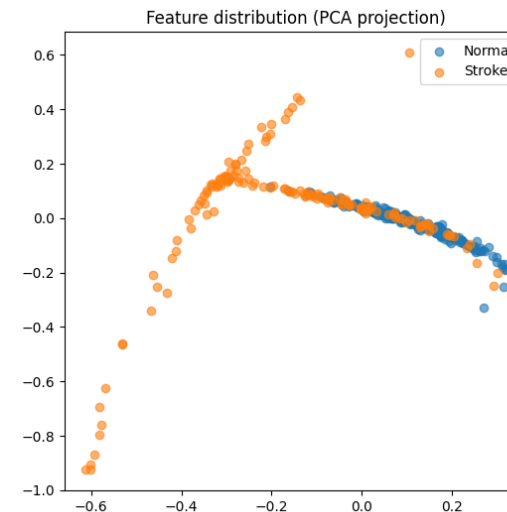
method	discriminant ratio
Shape Subspace	1.7002
Difference of Mean Vectors	2.2983
ICP	1.1033
Euclidean distance	2.3063



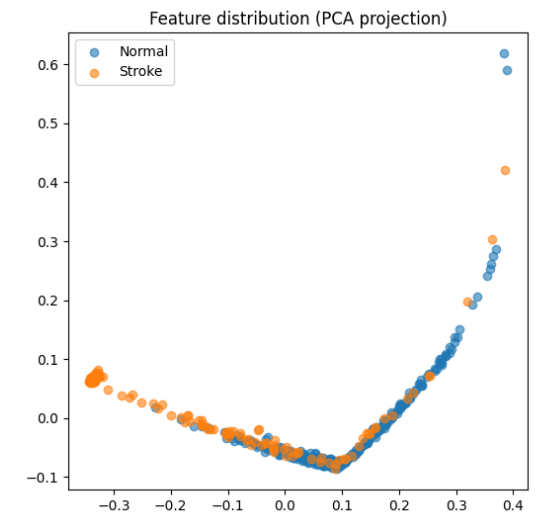
Shape Subspace



Difference of Mean Vectors



ICP



Euclidean distance

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■ Evaluation of Symmetry

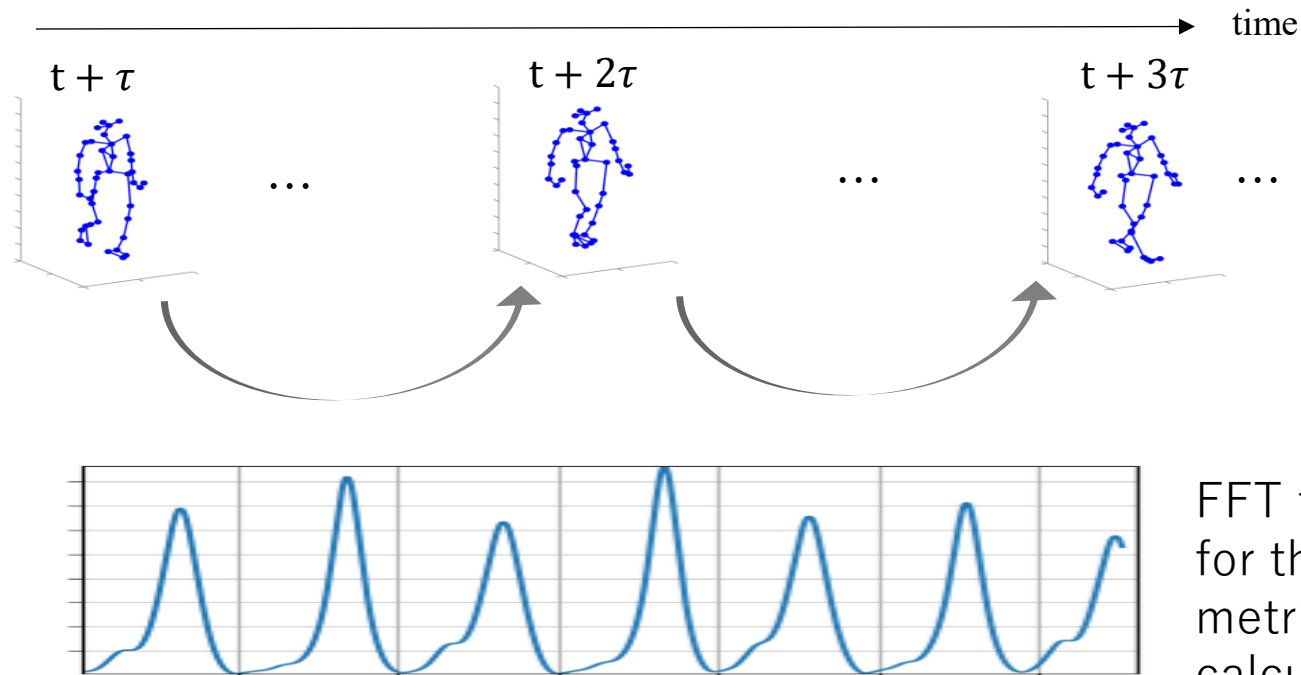
- Dimension of FFT features: 150 (half of 300 points covering 0 to 50 Hz)
- Training normal features 203
- Validation normal features 50
- Stroke All features 152

method	AUC(Dimension of the normal subspace: 5)	AUC(Dimension of the normal subspace: 3)	AUC(Dimension of the normal subspace: 2)
Shape Subspace	0.9099	0.9169	0.9267
Difference of Mean Vectors	0.5611	0.8167	0.8338
ICP	0.7153	0.8258	0.8404
Euclidean distance	0.6997	0.8591	0.8711

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■ Evaluation of Shape Variations

Instead of focusing on left-right shape differences, the evaluation focuses on overall temporal shape variations



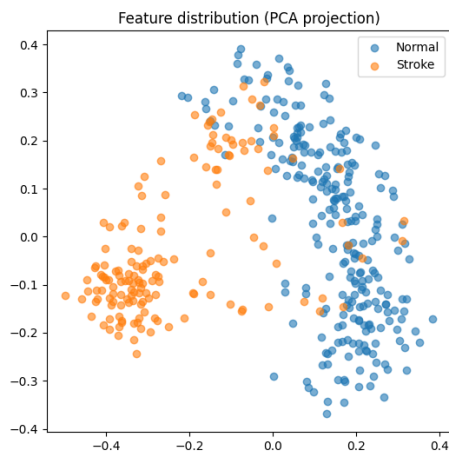
FFT features were computed for these variations, and metrics such as AUC were calculated in the same manner

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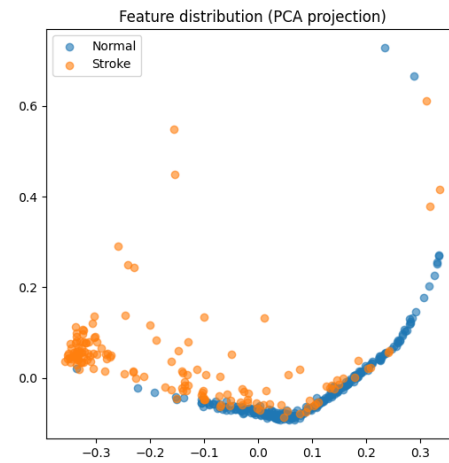
■ Evaluation of Shape Variations

- Distribution and separability (Fisher's discriminant ratio) of FFT features

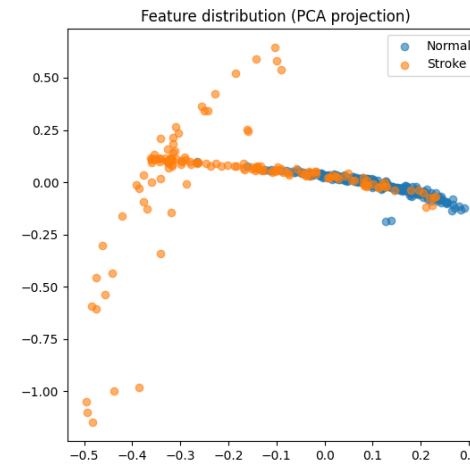
method	分離度
Shape Subspace	1.0664
Difference of Mean Vectors	1.5358
ICP	0.9511
Euclidean distance	1.4978



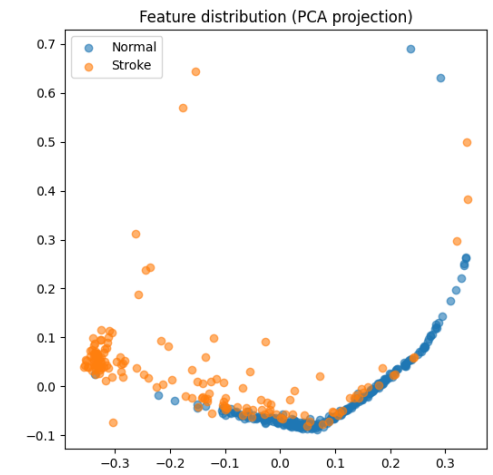
Shape Subspace



Difference of Mean Vectors



ICP



Euclidean distance

実験

■ Evaluation of Shape Variations

- Frame interval: 5
- Dimension of FFT features: 150 (half of 300 points covering 0 to 50 Hz)
- Training normal features 203
- Validation normal features 50
- Stroke All features 152

method	AUC(Dimension of the normal subspace: 5)	AUC(Dimension of the normal subspace: 3)	AUC(Dimension of the normal subspace: 2)
Shape Subspace	0.8968	0.9164	0.8949
Difference of Mean Vectors	0.9096	0.8863	0.9136
ICP	0.9036	0.9083	0.9092
Euclidean distance	0.9106	0.8866	0.9130

Feature Work / まとめ

- Gait evaluation considering both segmental coordination and temporal variations of the entire shape is possible
- Although gait data from patients with disorders are limited, stable analysis can be achieved using the subspace method, which is relatively robust to the amount of data
- In evaluating gait symmetry, the shape subspace method demonstrates particularly high effectiveness and may capture the geometric coordination between different body segments
- Changes in results due to hyperparameters and shape variations are also examined, including second-order differences
- We want to perform reliable gait evaluation